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Assignment: Assignment 01

Section: B

Course: Cloud Computing

Topic: Git, Gitea, GitHub, Git LFS, and GitHub Pages

1. Task 1 – Gitea

1.1 Gitea Installation

- Used Ubuntu Server as the working environment.
- Installed Docker and Docker Compose.
- Started the Gitea and PostgreSQL containers.
- Verified that both services were running successfully.
- Verified Gitea using the local HTTP request.
- Accessed Gitea through the Ubuntu server IP.
- Configured PostgreSQL as the database.
- Completed the Gitea installation and created the administrator account.

Figure 1 – Gitea running on Ubuntu Server

```
salehataj@ubuntu:~/Gitea$ docker compose ps
NAME                IMAGE                COMMAND                SERVICE    CREATED        STATUS        PORTS
gitea                gitea/gitea:latest  "/usr/bin/entrypoint..." gitea      8 seconds ago  Up 7 seconds  22/tcp, 0.0.0.0:3000->3000/tcp, [::]:3000->3000/tcp
gitea_db             postgres:17-alpine  "docker-entrypoint.s..." db         8 seconds ago  Up 7 seconds  5432/tcp
salehataj@ubuntu:~/Gitea$ curl -sS -o /dev/null -w "Gitea HTTP status: %{http_code}\n" http://127.0.0.1:3000/
Gitea HTTP status: 200
salehataj@ubuntu:~/Gitea$
```

1.2 Push Repository to Gitea

- Created a public Gitea repository named Assignment01.
- Created a local Git repository on Ubuntu.

- Initialized the repository with the main branch.
- Created README.md containing:
 - Full Name
 - Registration Number
- Committed the README file.
- Added Gitea as a remote named gitea.
- Successfully pushed the main branch to Gitea.

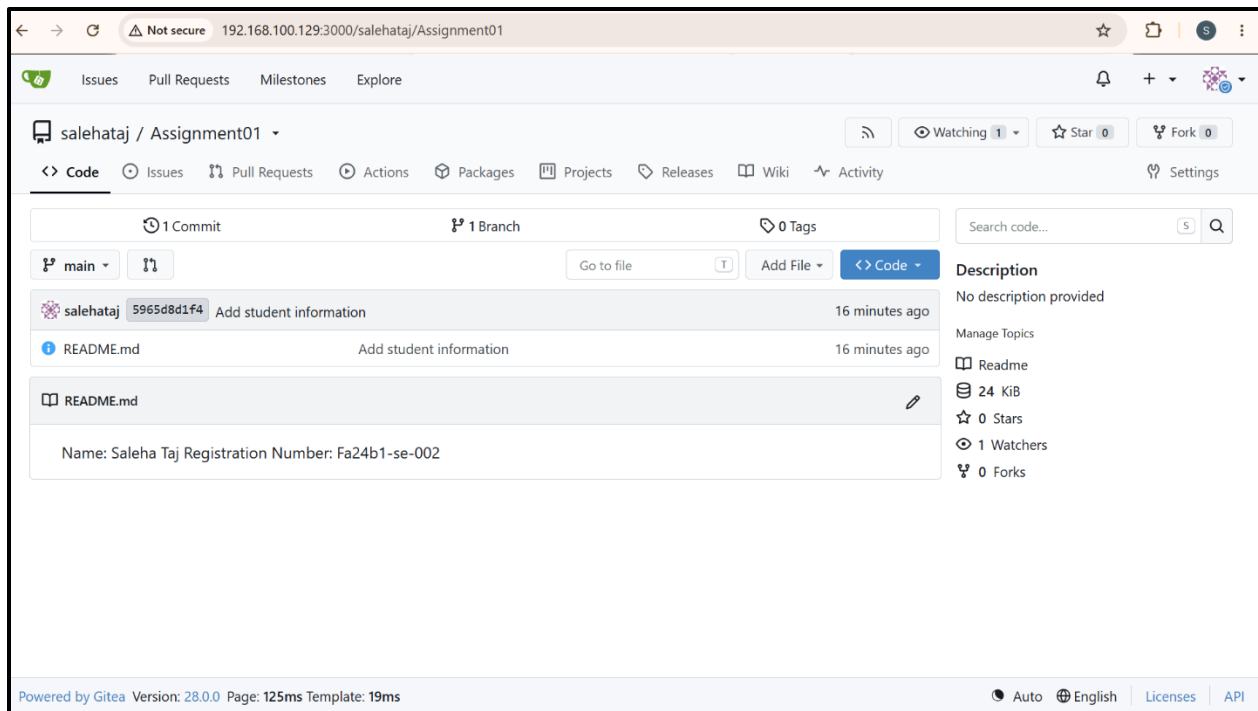
Figure 2 – Successful push to Gitea

```
salehataj@ubuntu:~/Assignment01$ git remote -v
gitea  http://salehataj@127.0.0.1:3000/salehataj/Assignment01.git (fetch)
gitea  http://salehataj@127.0.0.1:3000/salehataj/Assignment01.git (push)
salehataj@ubuntu:~/Assignment01$ git push -u gitea main
Password for 'http://salehataj@127.0.0.1:3000':
Enumerating objects: 3, done.
Counting objects: 100% (3/3), done.
Delta compression using up to 2 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (3/3), 273 bytes | 91.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To http://127.0.0.1:3000/salehataj/Assignment01.git
 * [new branch]      main -> main
branch 'main' set up to track 'gitea/main'.
salehataj@ubuntu:~/Assignment01$
```

1.3 Gitea Repository Verification

- Opened the Assignment01 repository in Gitea.
- Verified:
 - Repository name
 - Owner
 - Public visibility
 - README file
 - Student information

Figure 3 – Gitea Assignment01 repository



Result

- Gitea was successfully installed.
- Repository was successfully created and pushed.

2. Task 2 – GitHub

2.1 Configure Git Remotes

- Created a public GitHub repository named Assignment01.
- Used the same local repository from Task 1.
- Added GitHub as a second remote named github.
- Verified both remotes using `git remote -v`.
- The repository now had:
 - gitea remote
 - github remote

Figure 4 – Gitea and GitHub remotes

```
salehataj@ubuntu:~/Assignment01$ git remote -v
gitea http://salehataj@127.0.0.1:3000/salehataj/Assignment01.git (fetch)
gitea http://salehataj@127.0.0.1:3000/salehataj/Assignment01.git (push)
github https://github.com/salehataj/Assignment01.git (fetch)
github https://github.com/salehataj/Assignment01.git (push)
salehataj@ubuntu:~/Assignment01$
```

2.2 Push to GitHub

- Pushed the main branch to GitHub.
- GitHub authentication was completed successfully.
- Push completed without errors.

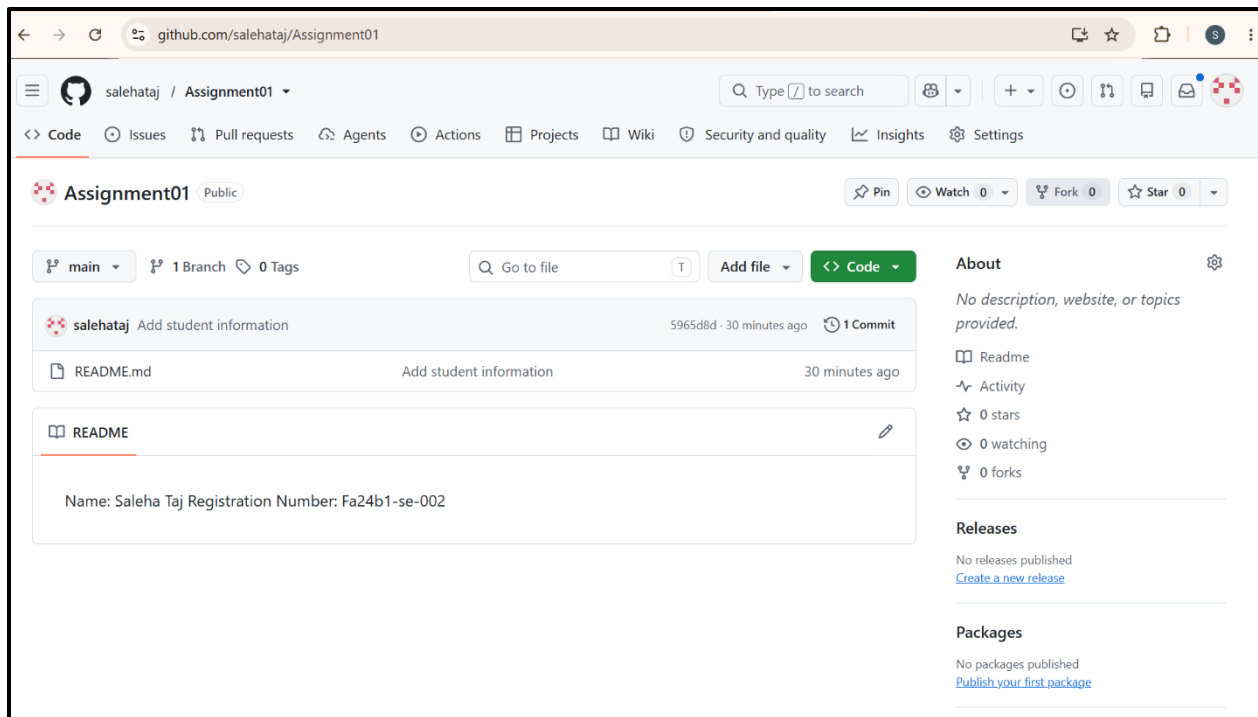
Figure 5 – Successful GitHub push

```
salehataj@ubuntu:~/Assignment01$ git push -u github main
Username for 'https://github.com': salehataj
Password for 'https://salehataj@github.com':
remote: Invalid username or token. Password authentication is not supported for Git operations.
fatal: Authentication failed for 'https://github.com/salehataj/Assignment01.git/'
salehataj@ubuntu:~/Assignment01$ git push -u github main
Username for 'https://github.com': salehataj
Password for 'https://salehataj@github.com':
Enumerating objects: 3, done.
Counting objects: 100% (3/3), done.
Delta compression using up to 2 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (3/3), 273 bytes | 273.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/salehataj/Assignment01.git
 * [new branch]      main -> main
branch 'main' set up to track 'github/main'.
salehataj@ubuntu:~/Assignment01$ _
```

2.3 GitHub Repository Verification

- Opened the GitHub Assignment01 repository.
- Verified:
 - Repository name
 - Owner
 - Public visibility
 - README file
 - Student information

Figure 6 – GitHub Assignment01 repository



Result

- The same local repository was successfully pushed to both Gitea and GitHub.

3. Task 3 – Git LFS

3.1 Git LFS Setup

- Initialized Git LFS in the Assignment01 repository.
- Verified the installed Git LFS version.
- Configured .bin files to be tracked using Git LFS.
- Verified the .gitattributes file.

Figure 7 – Git LFS setup

```
salehataj@ubuntu:~/Assignment01$ git lfs install
Updated Git hooks.
Git LFS initialized.
salehataj@ubuntu:~/Assignment01$ git lfs version
git-lfs/3.7.1 (GitHub; linux amd64; go 1.25.0)
salehataj@ubuntu:~/Assignment01$ git lfs track "*.bin"
Tracking "*.bin"
salehataj@ubuntu:~/Assignment01$ cat .gitattributes
*.bin filter=lfs diff=lfs merge=lfs -text
salehataj@ubuntu:~/Assignment01$ _
```

3.2 Large File Creation

- Created three binary files:

- large-file-1.bin – 101 MiB
- large-file-2.bin – 102 MiB
- large-file-3.bin – 103 MiB
- Added the files to Git.
- Verified that all three files were tracked by Git LFS.

Figure 8 – Git LFS tracked files

```
salehataj@ubuntu:~/Assignment01$ truncate -s 101M large-file-1.bin
salehataj@ubuntu:~/Assignment01$ truncate -s 102M large-file-2.bin
salehataj@ubuntu:~/Assignment01$ truncate -s 103M large-file-3.bin
salehataj@ubuntu:~/Assignment01$ git add .gitattributes large-file-1.bin large-file-2.bin large-file-3.bin
salehataj@ubuntu:~/Assignment01$ git lfs ls-files --size
f1753a034b * large-file-1.bin (106 MB)
2338a970c0 * large-file-2.bin (107 MB)
cb91d5908e * large-file-3.bin (108 MB)
salehataj@ubuntu:~/Assignment01$
```

3.3 Git LFS Push

- Committed the three large files.
- Pushed the commit to GitHub.
- Git LFS uploaded the large file objects successfully.
- Push completed successfully.

Figure 9 – Successful Git LFS push

```
salehataj@ubuntu:~/Assignment01$ git push github main
Username for 'https://github.com': salehataj
Password for 'https://salehataj@github.com':
Uploading LFS objects: 100% (3/3), 321 MB | 179 KB/s, done.
Enumerating objects: 7, done.
Counting objects: 100% (7/7), done.
Delta compression using up to 2 threads
Compressing objects: 100% (5/5), done.
Writing objects: 100% (6/6), 793 bytes | 264.00 KiB/s, done.
Total 6 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/salehataj/Assignment01.git
 5965d8d..e4e7b1e main -> main
salehataj@ubuntu:~/Assignment01$
```

Result

- Three files larger than 100 MiB were successfully managed and pushed using Git LFS.

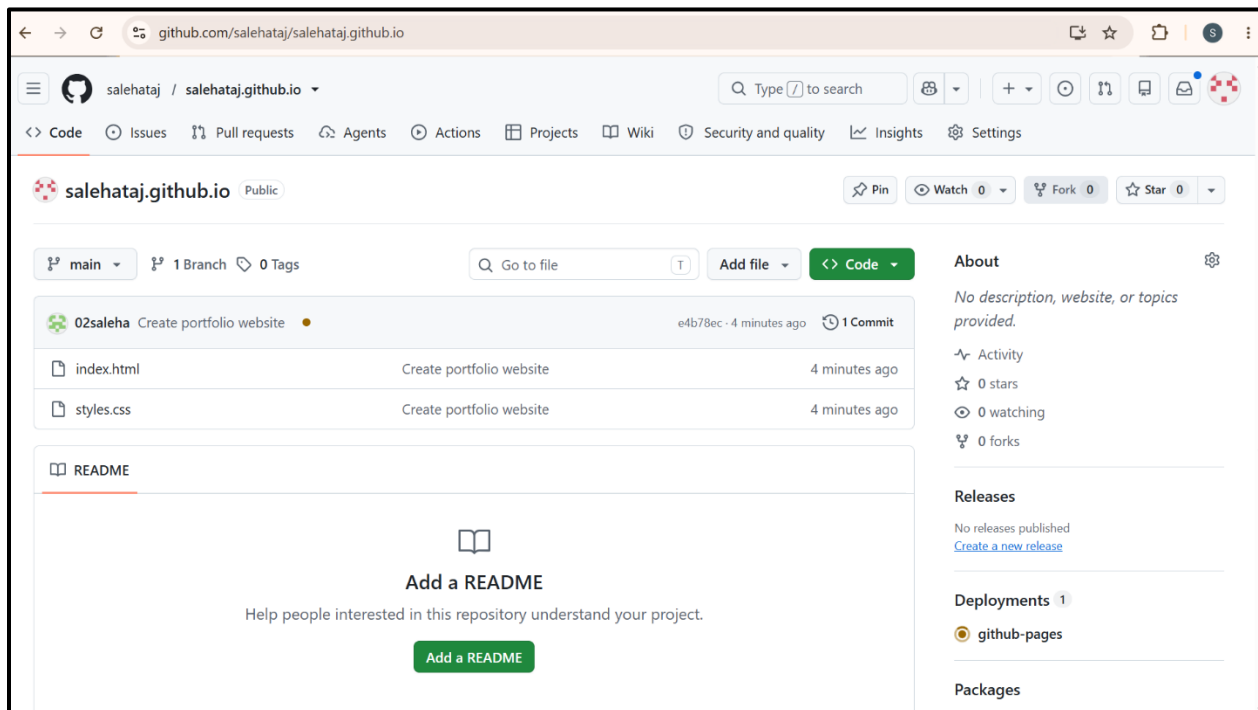
4. Task 4 – GitHub Pages Portfolio

4.1 Portfolio Creation

- Created a personal portfolio website using HTML and CSS.

- Created the following files:
 - index.html
 - styles.css
- Added the following sections:
 - About Me
 - Education
 - Skills
 - Projects
- Used a green, white, and black design.
- Created a GitHub repository named salehataj.github.io.
- Pushed the portfolio to the main branch.

Figure 10 – Portfolio GitHub repository



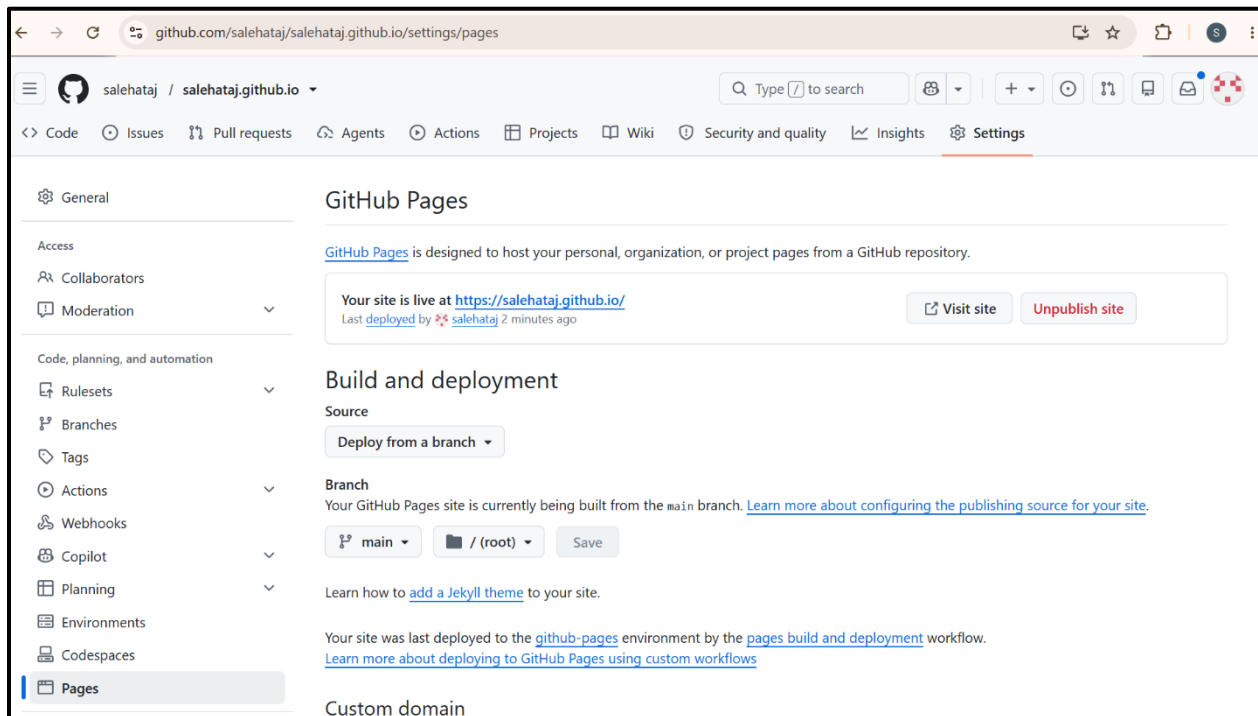
4.2 GitHub Pages Deployment

- Opened the repository's **Settings** → **Pages**.

- Selected deployment from the main branch.
- Selected the root directory / (root).
- Saved the configuration.
- GitHub Pages successfully deployed the website.
- Generated live website URL:

<https://salehataj.github.io/>

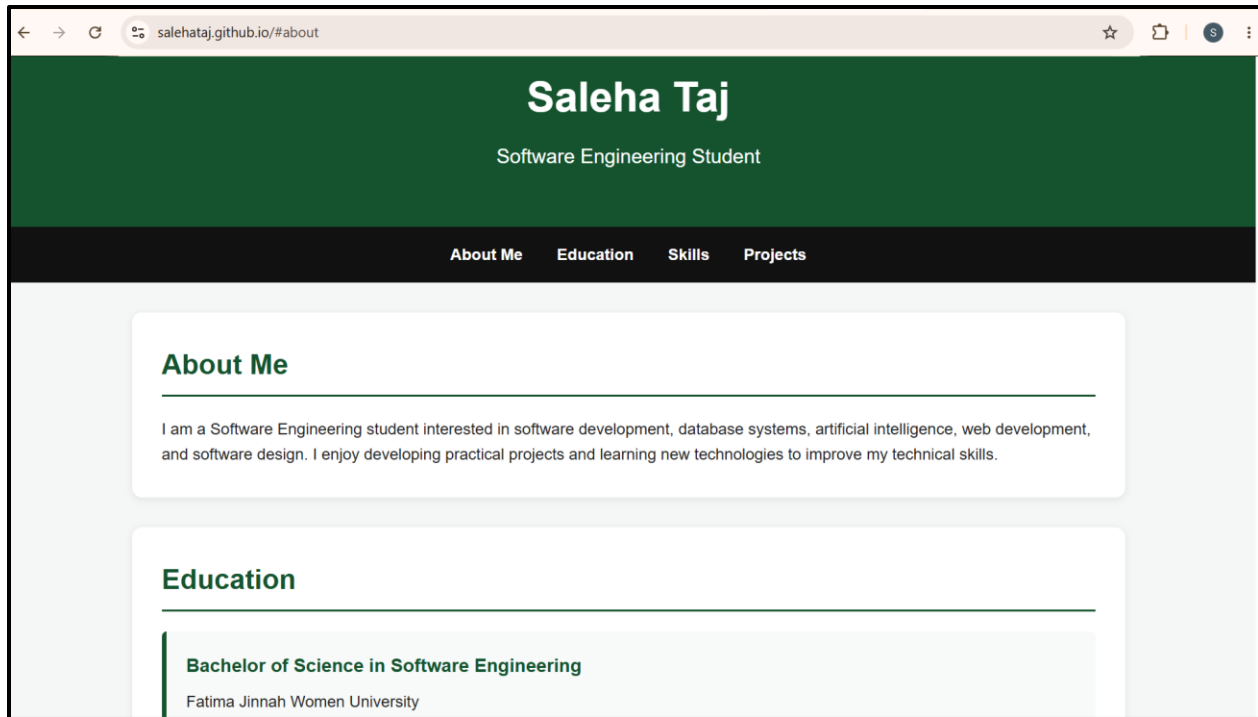
Figure 11 – GitHub Pages deployment



4.3 Live Portfolio

- Opened the GitHub Pages URL in the browser.
- Verified that the website was publicly accessible.
- Confirmed that the portfolio content was displayed correctly.

Figure 12 – Live portfolio website



Result

- The portfolio website was successfully created and published using GitHub Pages.